Synchronous speed, N_s= rpm

Rated speed = N = rpm

Slip , s = $\frac{N_s - N}{N_s} \times 100\% =$ ______%

$R_L' = R_2'\left(\frac{1-s}{s}\right) =$ ______ Ω

$\overline{I_0} = I_0 < -\emptyset_0° =$ ______ A *per phase*

From approximate equvalent circuit,

$\overline{I_2'} = \frac{V<0^0}{(R_{01} + R_L) + jX_{01}} =$ ______ A *per phase*

$\overline{I_1} = \overline{I_0} + \overline{I_2'} = I_1 < -\emptyset_1° =$ ______ A

Line current $I_L = \sqrt{3}I_1 =$ ______ A

Power factor = $\cos(-\emptyset_1°) =$ ______

Output = $3 \times I_2'^2 \times R_L' =$ ______ W

Torque $= \frac{output}{(2\pi N/60)} =$ ______ Nm

Input = $\sqrt{3}VI_L \cos\emptyset_1 =$ ______ W

Efficiency $= \frac{output}{input} \times 100\% =$ ______%

RESULT

- No-load and blocked rotor tests were conducted on 3 phase squirrel cage induction motor
- Equivalent circuit parameters were determined
- Circle diagram was drawn
- Performance at full load from equivalent circuit and circle diagram were determined